The Evolution of Web Protocols

In this new series of articles, we take a deep dive into Web protocols and look at how these have evolved over time. This article is a must for anyone who wishes to understand the Internet and the way it works.

The Internet is one of the biggest technological marvels of modern times, enabling easy communication across the globe, opening up new ways of commerce and supporting a wide variety of other conveniences. While it is referred to in the singular, it is composed of a plethora of hardware and software solutions. Multiple governments, communities and corporations have, individually and collaboratively, made huge investments to build the infrastructure and applications of the Internet.

Overall, the network consists of the application (user-visible) and infrastructural layers. Given the scale of the Internet, both comprise a humongous number of software and hardware components. We look at a few examples of these in Table 1 to get a feel of the variety of components deployed.

Among the entities in this table, our interest lies in the software components, and particularly on those highlighted in bold text. Most of the traffic driven by these elements belongs to a handful of protocols, dominant among which are DNS, HTTP, SSL/TLS, TCP and IP — these are literally the workhorses of the Internet. These protocols are layered as a stack, which is described in detail in the next section.

Our interest in this series of articles is to understand how each protocol evolved individually and as part of the stack, and the reasons that led to the evolution. The series is structured as follows.

In this first article in the series, we start with outlining the end-to-end flow of information for Web interactions. Then, we step back in time and mark the key milestones in the evolution of the protocol stack. We also dive into the early protocols including HTTP 1.1, DNS, SSL/TLS, and OCSP. This forms the basis for understanding the newer protocols.

Table 1

	Hardware	Software
Network infrastructure – wired	Switches, routers, firewalls, load balancers, electronic and optical physical layer systems, cables, and so on	**Protocol stacks, CDNs, security gateways**
Network infrastructure - mobile-specific	Antennas, base stations, and so on	2G, 3G, LTE and 5G stacks
Network applications	Mobile phones, desktops, servers and other such devices	**Browsers, Web servers, auxiliary software (CAs, OCSP)**